

Impacts of Artificial Intelligence

KEDGE - Data Analytics for Business

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Today

- Work on text + video
- Discussion time + additional elements from the article of TUBARO P., CASILLI A.A., COVILLE M.
- (More) Time to work on the final presentation

Session 3: AI and the transformation of organizations and work

Reminder

Session 3: AI and the transformation of organizations and work

- GRAY M.L., SURI S., 2017, « The Humans Working Behind the AI Curtain », [*Harvard Business Review*](#), 9 janvier 2017.
- *Complementary reading:* CASILLI A.A., 2021, « Waiting for robots: the ever-elusive myth of automation and the global exploitation of digital labor », [*Sociologias*](#), 23, 57, p. 112 133.
- *Complementary reading:* TUBARO P., CASILLI A.A., COVILLE M., 2020, « The Trainer, the Verifier, the Imitator: Three Ways in Which Human Platform Workers Support Artificial Intelligence », [*Big Data & Society*](#), 7, 1, p. 1 12.

Casilli A.A., 2019, En attendant les robots

Enquête sur le travail du clic

Antonio A. Casilli
EN ATTENDANT
LES ROBOTS
Enquête sur le travail du clic



POURQUOI DES ROBOTS ?
SEUIL

Automation

Humans replacing robots

Platforms

Digital labor

On-demand labor

Microwork

Networked labor

Casilli



Professor of sociology

- telecommunication school (Telecom Paris) of the Institute Polytechnique de Paris
- researcher at the Interdisciplinary Institute on Innovation (i3)

Research areas

- social networks, digital platforms, digital labor and privacy

Digital labor

- On-demand labor
 - Uber, Lyft, Airbnb, Taskrabbit, Glovo and Deliveroo that provide logistic, transportation, and care services
- Microwork
 - small repetitive tasks outsourced to crowds of independent providers
- Networked labor
 - “activities that are indispensable to run social media and generic user-based platforms, such as Google, Facebook, Instagram, YouTube. Despite the emphasis that these companies have put on the specialized work of computer scientists, engineers, and code developers, social networked labor does not relate to these elite workers, but rather to unrecognized contributors – either precarious clickworkers or unpaid generators of content and data”

EXERCISE

EXERCISE - Text & video

- Watch the [documentary](#)
- Read the text GRAY M.L., SURI S., 2017, « The Humans Working Behind the AI Curtain », [*Harvard Business Review*](#), 9 janvier 2017.

Interpretation grid

- what roles for humans in AI?
- working conditions?
- etc.

Discussion time

**Sharing of results, confrontation
of ideas, discussion and
exchange**

More elements

Human Workers and Artificial Intelligence

Tubaro P., Casilli A.A., Coville M., 2020,
« The trainer, the verifier, the imitator:
Three ways in which human platform
workers support artificial intelligence »,
Big Data & Society, 7, 1, p. 1-12.

The trainer

The verifier

The imitator

Main ideas

Micro-work

- ‘artificial intelligence preparation’, ‘artificial intelligence verification’
‘artificial intelligence impersonation’

Structural ! not ephemeral

- “Because of the wide scope of application of micro-work, it is a structural component of contemporary artificial intelligence production processes – not an ephemeral form of support that may vanish once the technology reaches maturity stage.”

Fieldwork

Mixed-methods, technology companies, 2017–2018 in France

- “day-to-day routines of platform workers” that took place in 2 (Casilli et al., 2019).
- something different than classical Amazon Mechanical Turk (specific to the US)
- “trend toward diversification and specialization of micro-work platforms, attracting an ever-wide range of users.”

3 stakeholders groups

Triangulation of information between AI producers / microwork platforms / people who perform micro-tasks

AI producers

- ++ “best understanding of technology and its needs”
- -- “unwilling or unable to disclose their use of human workers”

Platforms

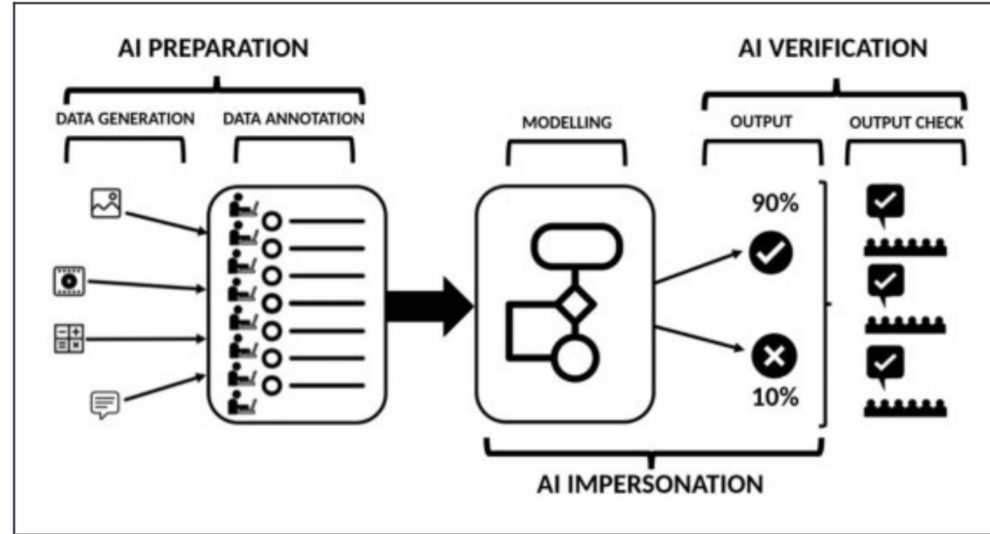
- ++ “best positioned to know the structure of the market”
- -- “need to attract clients and investors may inflect their communication strategies”.

Workers

- ++ “share the unique, concrete experience of doing micro-tasks”
- -- “not always aware of their purposes and final uses”

Micro-work & AI data preparation... & more

- “AI companies depend heavily on data resources, including **not only raw data** but also **annotations**”
- “They are **challenges** for AI producers, **despite a widespread rhetoric of ‘data deluge’** (Anderson, 2008): the **right data** are not always available or accessible, and when they are, they often lack suitable annotations, and need intervention before they can be used.”



The trainer - AI preparation

Micro-works and data preparation / annotation

AI in the loop ? Sometimes, help from AI during this process, BUT

- assist human and accelerate the pace of work = human remains essential
- “not eliminated the need for micro-tasking, but transformed it, integrating humans and computers more tightly”.

DeepLearning development increases demand

- Lukas Biewald (Figure Eight Inc) : need for larger labelled datasets : “Deep learning has been fantastic [. . .] We began noticing deep learning when we started having customers who would ask for tens of millions of data rows right off the bat.” p. 6

Micro-workers' experience of AI preparation

- confrontation with the absence of meaning (or understanding of why the request is made)
- Confrontation with violent content, porn, etc. (content moderation)

The imitator - AI impersonation

Reminder (cf. Crawford)

- start-ups advertised as AI but not doing AI = communication, make believe, cheat, while humans are doing the work behind the curtain, invisibilized

Impersonation

- “when humans, [...] steal computers’ jobs”. » Amazon Mechanical Turk
- Impersonation is *not just about fraud*, and indeed Amazon has always been upfront about it. It is the ‘human-in-the-loop’ principle that makes workers *hardly distinguishable from algorithms*. [...] More generally, the idea is that *whenever an algorithm cannot autonomously bring an activity to completion, it hands control over to a human operator*. p. 7
- “Micro-work platforms do provide human labour force to meet these needs, but **tend not to explicitly advertise these roles**: they arguably negotiate impersonating tasks individually with clients as part of their managed service packages. Humans are always in the loop, but they are **even less visible here**, than when they do ‘preparation’ tasks.” p. 8

The verifier - AI verification

Controversy over the listening of voice assistant recordings by humans.

Common points with *AI impersonation*, but difference comes from the fact that it **does not replace a failing algorithm**: checks reliability/quality.

Common points with *AI preparation*, but **intervenes after** the algorithm

- "she was intervening 'post-robot': she produced training data from the amended outputs of an already-trained algorithm"

Examples of tasks located after preparation phase: relevant scoring, checking of optical character recognition (OCR) results.

There will always be a need to verify the quality of the algorithms so this type of work is also **here to stay**.

Discussion and conclusions

The contribution of micro-workers is not temporary but structural.

Interesting point about the availability of datasets and their production/quality which will always be necessary:

- “Data availability will never reach a steady state: most use cases for machine learning require ongoing acquisition of new sources to continuously adjust to changing conditions, resulting in a steadily growing need for humans to produce data for more accurate, more precise, and more profitable results.”

Mention the possible overlap of their categories + the fact that some activities may not fit into these categories (click farms).

Issue of micro-laborer working conditions

Group work on final presentation

The Goals

- Stable groups
- Choice of topic
- Started collecting resources and working on the topic
- Exchange with me to guide you according to your needs

Evaluation = Oral presentation

- On a theme of your choice related to the course (if justified, a case study can be done)
- In **groups of ~5** depending on the number of participants): ideally around 6 groups
- **~15 minutes** (depending on the number of groups)

Expectations / guidelines - 1

Gather documentation:

- At least **two** articles from the national/international **press**
 - French examples: Le Figaro, Le Monde, Libération, etc.
 - International examples: NYT, The Washington Post, El Pais, etc.
- At least **one** article from **academic** journals that will help you organize your reasoning
- At least **one book** (no obligation to read it in its entirety, but at least rely on it + you can consult reading notes in academic journals or social science sites)
- Also possible (no obligation): statistics (INSEE, Eurostat, etc.), radio programs / podcasts

Expectations / guidelines - 2

Construct a problematic

- Propose a presentation that is not a juxtaposition of descriptive elements but a rigorous and coherent argumentation allowing to answer the initial question.

Format: PowerPoint (or other presentation tools)

- Propose a presentation based on a rich and neat PowerPoint. Write down the essential arguments
- Insert a few significant quotations if necessary
- Bibliography : specify at the end of the PowerPoint the documentation used

Duration

- Each presentation lasts 15-20 minutes and is followed by a 10-minute discussion.